

Lab Tasks

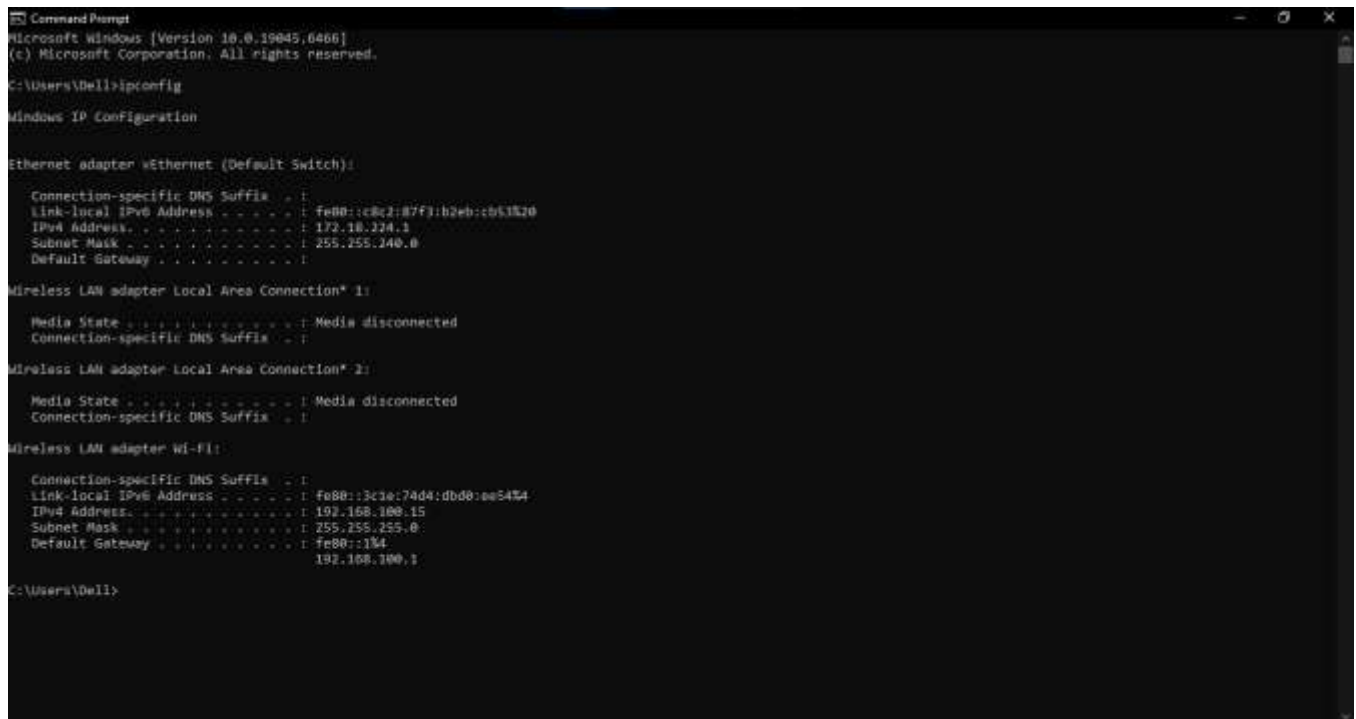
(Attach screenshot of command prompt/terminal wherever required.)

(AI-generated answers are not acceptable. Make sure all responses are written in your own words.)

- 1) Find the IP address of the computer you are currently using.

Command: ipconfig

IP Address: 192.168.100.15

A screenshot of a Windows Command Prompt window. The title bar says "Command Prompt". The text inside shows the command 'ipconfig' being executed. The output displays network configuration for three adapters: Ethernet adapter Ethernet (Default Switch), Wireless LAN adapter Local Area Connection* 1, and Wireless LAN adapter Local Area Connection* 2. The Ethernet adapter is active and shows an IPv4 address of 172.18.224.1. The two wireless adapters are disconnected. A third section, 'Wireless LAN adapter Wi-Fi', is also shown with an IPv4 address of 192.168.100.15.

```
Microsoft Windows [Version 10.0.19045.6466]
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C:\Users\Dell>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet (Default Switch):

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::c8c2:b2eb:c953%20
    IPv4 Address. . . . . : 172.18.224.1
    Subnet Mask . . . . . : 255.255.240.0
    Default Gateway . . . . . : 

Wireless LAN adapter Local Area Connection* 1:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Local Area Connection* 2:

    Media State . . . . . : Media disconnected
    Connection-specific DNS Suffix . : 

Wireless LAN adapter Wi-Fi:

    Connection-specific DNS Suffix . : 
    Link-local IPv6 Address . . . . . : fe80::3c3e:74d4:dbd8:ae84%4
    IPv4 Address. . . . . : 192.168.100.15
    Subnet Mask . . . . . : 255.255.255.0
    Default Gateway . . . . . : fe80::1%4
                                192.168.100.1

C:\Users\Dell>
```

- 2) Find the IP address of the computer you are currently using, plus MAC address, the gateway, plus whether DHCP is turned on.

Command: ipconfig /all

IP Address: 192.168.100.15

MAC Address: 00-15-5D-E5-A9-13

Gateway: fe80::1%4
192.168.100.1

DHCP: Turned On

```
Command Prompt
C:\Users\ DELL\ipconfig /all

Windows IP Configuration

Host Name . . . . . : DESKTOP-H3K63R6
Primary Dns Suffix . . . . . :
Node Type . . . . . : Mixed
IP Routing Enabled. . . . . : No
WINS Proxy Enabled. . . . . : No

Ethernet adapter vEthernet (Default Switch):

Connection-specific DNS Suffix . :
Description . . . . . : Hyper-V Virtual Ethernet Adapter
Physical Address. . . . . : 08-15-50-E5-A9-13
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::c8c2:87f3:b2eb:c533%10(Preferred)
IPv4 Address. . . . . : 172.18.224.1(Preferred)
Subnet Mask . . . . . : 255.255.240.0
Default Gateway . . . . . :
DHCPv6 IAID . . . . . : 335549780
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-80-70-83-84-05-BD-2B-DF-C5
DNS Servers . . . . . : fec8:0:0:ffff::1%1
                        fec8:0:0:ffff::2%1
                        fec8:0:0:ffff::3%1
NetBIOS over Tcpip. . . . . : Enabled

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : B4-06-80-2B-DF-C6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : B6-06-80-2B-DF-C5

Command Prompt

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : B4-06-80-2B-DF-C6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . :
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : D6-06-80-2B-DF-C5
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description . . . . . : Intel(R) Dual Band Wireless-AC 7265
Physical Address. . . . . : B4-06-80-2B-DF-C5
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3c1e:74d4:dhd0:ee54%4(Preferred)
IPv4 Address. . . . . : 192.168.100.15(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, January 25, 2016 10:02:27 PM
Lease Expires . . . . . : Monday, January 26, 2016 10:02:29 PM
Default Gateway . . . . . : fe80::1%4
                        192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DHCPv6 IAID . . . . . : 62182845
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-80-70-83-84-05-BD-2B-DF-C5
DNS Servers . . . . . : 192.168.100.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\ DELL>
```

3) Display the host name of the computer.

Command: hostname

Name: DESKTOP-H3K63R6

```
Command Prompt

Wireless LAN adapter Local Area Connection* 1:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. . . . . : B4-05-80-2B-DF-C6
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Local Area Connection* 2:

Media State . . . . . : Media disconnected
Connection-specific DNS Suffix . : 
Description . . . . . : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. . . . . : 06-05-80-2B-DF-C5
DHCP Enabled. . . . . : No
Autoconfiguration Enabled . . . . : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . : 
Description . . . . . : Intel(R) Dual Band Wireless-AC 7265
Physical Address. . . . . : B4-05-80-2B-DF-C5
DHCP Enabled. . . . . : Yes
Autoconfiguration Enabled . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3c1e-74d4-dbd0-ee54%4(Preferred)
IPv4 Address. . . . . : 192.168.100.15(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, January 25, 2016 10:02:27 PM
Lease Expires . . . . . : Monday, January 26, 2016 10:02:29 PM
Default Gateway . . . . . : fe80::1%4
                          : 192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DHCPv6 IAID . . . . . : 02182845
DHCPv6 Client DUID. . . . . : 00-01-00-01-2A-60-70-81-B4-05-80-2B-DF-C5
DNS Servers . . . . . : 192.168.100.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\Dell>hostname
DESKTOP-H3K03RQ

C:\Users\Dell>
```

4) Check for basic IP connectivity between two computers by name and IP address. How can basic IP connectivity be checked? What are the reasons why there is no connectivity?

Command: ping

How: Basic connectivity is checked by sending ICMP "echo request" packets to the target. If the target receives them, it sends a reply.

Reason:

- **Firewall:** The destination computer's firewall might be blocking ICMP packets (even if the connection is actually good).
- **Physical Issue:** A cable might be unplugged or the Wi-Fi disconnected.
- **Wrong Network:** The computers might be on different subnets without a router to bridge them.

```
Command Prompt
Autoconfiguration Enabled . . . . . : Yes
Link-local IPv6 Address . . . . . : fe80::3c1e:74d4:dbd0:ee54M4(Preferred)
IPv4 Address. . . . . : 192.168.100.15(Preferred)
Subnet Mask . . . . . : 255.255.255.0
Lease Obtained. . . . . : Sunday, January 25, 2016 10:02:27 PM
Lease Expires . . . . . : Monday, January 26, 2016 10:02:29 PM
Default Gateway . . . . . : fe80::1154
                          : 192.168.100.1
DHCP Server . . . . . : 192.168.100.1
DHCPv6 IAID . . . . . : 02182845
DHCPv6 Client GUID. . . . . : 00-01-00-01-20-69-70-83-04-05-BD-28-DF-C5
DNS Servers . . . . . : 192.168.100.1
NetBIOS over Tcpip. . . . . : Enabled

C:\Users\De11>hostname
DESKTOP-H3K0386

C:\Users\De11>ping google.com

Pinging google.com [142.250.203.174] with 32 bytes of data:
Reply from 142.250.203.174: bytes=32 time=20ms TTL=115
Reply from 142.250.203.174: bytes=32 time=26ms TTL=115
Reply from 142.250.203.174: bytes=32 time=27ms TTL=115
Reply from 142.250.203.174: bytes=32 time=26ms TTL=115

Ping statistics for 142.250.203.174:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 20ms, Maximum = 27ms, Average = 24ms

C:\Users\De11>ping www.google.com

Pinging www.google.com [142.250.202.4] with 32 bytes of data:
Reply from 142.250.202.4: bytes=32 time=21ms TTL=115
Reply from 142.250.202.4: bytes=32 time=21ms TTL=115
Reply from 142.250.202.4: bytes=32 time=21ms TTL=115
Reply from 142.250.202.4: bytes=32 time=21ms TTL=115

Ping statistics for 142.250.202.4:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 21ms, Average = 21ms

C:\Users\De11>
```

- 5) Find out which ports on your host are connected to applications. Connect the browser to some external web page before running the appropriate command.

Command: netstat -a


```
Command Prompt
UDP 0.0.0.0:51667 *:*
UDP 0.0.0.0:57778 *:*
UDP 0.0.0.0:57961 *:*
UDP 0.0.0.0:59132 *:*
UDP 0.0.0.0:63614 *:*
UDP 0.0.0.0:64326 *:*
UDP 0.0.0.0:65391 *:*
UDP 127.0.0.1:1900 *:*
UDP 127.0.0.1:49664 *:*
UDP 127.0.0.1:49937 *:*
UDP 172.18.224.1:07 *:*
UDP 172.18.224.1:68 *:*
UDP 172.18.224.1:137 *:*
UDP 172.18.224.1:138 *:*
UDP 172.18.224.1:1900 *:*
UDP 172.18.224.1:49935 *:*
UDP 192.168.100.15:137 *:*
UDP 192.168.100.15:138 *:*
UDP 192.168.100.15:1900 *:*
UDP 192.168.100.15:49936 *:*
UDP [::]:3702 *:*
UDP [::]:3702 *:*
UDP [::]:3702 *:*
UDP [::]:3702 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5355 *:*
UDP [::]:5355 *:*
UDP [::]:56580 *:*
UDP [::]:57962 *:*
UDP [::]:64127 *:*
UDP [::]:1900 *:*
UDP [::]:49934 *:*
UDP [::]:58012 *:*
UDP [fe80::3c1e:74d4:dbd0:ee54%4]:1900 *:*
UDP [fe80::3c1e:74d4:dbd0:ee54%4]:49933 *:*
UDP [fe80::c8c2:87f3:b2eb:cb53%20]:1900 *:*
UDP [fe80::c8c2:87f3:b2eb:cb53%20]:49932 *:*
C:\Users\De11>
```

6) Find the path of routers to www.google.com. What is its IP address? How many hops involved in the path?

Command: `tracert www.google.com`

Reason: This command traces the route packets take to reach Google.

- **IP Address:** 142.250.202.4
- **Hops:** 13

```
Command Prompt
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5353 *:*
UDP [::]:5355 *:*
UDP [::]:50580 *:*
UDP [::]:57062 *:*
UDP [::]:64327 *:*
UDP [::]:1900 *:*
UDP [::]:49934 *:*
UDP [::]:50012 *:*
UDP [fe80::3c1e:74d4:dbd0:ee5454]:1000 *:*
UDP [fe80::3c1e:74d4:dbd0:ee5454]:40033 *:*
UDP [fe80::c8c2:87f3:b2eb:cb53320]:1000 *:*
UDP [fe80::c8c2:87f3:b2eb:cb53320]:49932 *:*

C:\Users\Oell>tracert www.google.com

Tracing route to www.google.com [142.250.202.4]
over a maximum of 30 hops:
  0  0 ms  0 ms  0 ms  192.168.100.1
  1  10 ms  11 ms  9 ms  100.87.0.1
  2  *  *  *  Request timed out.
  3  *  *  *  Request timed out.
  4  *  *  *  Request timed out.
  5  *  *  *  Request timed out.
  6  *  *  *  Request timed out.
  7  *  *  *  Request timed out.
  8  *  *  *  Request timed out.
  9  *  *  *  Request timed out.
 10  *  *  *  Request timed out.
 11  *  *  *  Request timed out.
 12  *  *  *  Request timed out.
 13  21 ms  25 ms  20 ms  poxfra-af-in-f4.1e100.net [142.250.202.4]

Trace complete.

C:\Users\Oell>
```

7) A ping to 192.168.0.2 works but a ping to the machine's name "blue machine" fails. What could be wrong?

Reason: This is a DNS (Domain Name System) or name resolution issue.

- The network connection itself is fine (proven because the IP ping worked).
- However, the computer does not know how to translate the name "blue machine" into the IP address 192.168.0.2. This happens if the DNS server is down or if there is no entry for "blue machine" in the local hosts file.

8) Which type of cable will you use to connect in a normal home installation?

Answer: For a normal home installation, we use Unshielded Twisted Pair (UTP) cable. The specific standard is usually Cat5e or Cat6 with RJ45 connectors.

9) Can you connect a Switch to another Switch or a router to a PC using a straight- through cable? Explain your answer.

Answer: Technically, we are supposed to use a Crossover cable, not a straight-through cable because Switches (or Routers and PCs) are "like" devices that transmit on the same pins. If you use a straight-through cable, the "send" pins connect to the other side's "send" pins, causing a collision.

10) Write a brief report on your home network or any organizational network including topology, 1 page max).

Answer:

My Home Network Report: My home network utilizes a Star Topology. The central node is a wireless router provided by the ISP (fiber connection).

- **Hardware:** 1 Fiber Modem/Router Combo (Gateway), 3 Smartphones, 2 Laptops, and 1 Smart TV.
- **Connection:** The Smart TV is connected via a wired Cat6 LAN cable for stability. All other devices (phones and laptops) connect via Wi-Fi (802.11ac).
- **IP Assignment:** The router acts as the DHCP server, assigning private IP addresses (Range 192.168.1.x) to all devices.